2. A teacher wants to arrange student marks in ascending order and also measure how much

rearrangement is necessary. Write a C program using Insertion Sort that accepts n marks, displays the array after every pass, counts the total number of element shifts, and displays the final sorted list and shift count.

Sourcecode :-

```
#include <stdio.h>

int main()

{

   int a[100], n, i, j, key;

   int shifts = 0;

   printf("Enter number of students: ");

   scanf("%d", &n);

   printf("Enter student marks:\n");

   for (i = 0; i < n; i++)

      scanf("%d", &a[i]);

   for (i = 1; i < n; i++)
```

```
{
        key = a[i];
        j = i - 1;
        while (j >= 0 && a[j] > key)
        {
            a[j + 1] = a[j];
            j--;
            shifts++;
        }
        a[j + 1] = key;
        printf("Pass %d: ", i);
        for (int k = 0; k < n; k++)
        {
            printf("%d ", a[k]);
        }
        printf("\n");
    }
    printf("\nFinal sorted list: ");
    for (i = 0; i < n; i++)
    {
        printf("%d ", a[i]);
    }
    printf("\nTotal number of shifts: %d\n", shifts);
    return 0;
}
```

Output :-

```
Enter number of students: 5
Enter student marks:
100 99 97 98 95
Pass 1: 99 100 97 98 95
Pass 2: 97 99 100 98 95
Pass 3: 97 98 99 100 95
Pass 4: 95 97 98 99 100

Final sorted list: 95 97 98 99 100
Total number of shifts: 9
```